Evolution 0: Course Self Assessment

Course (number and title)	Knowledge: what you know	Skills and tools: how you use it	Strongest (3–5)	Weakest (3–5)	GitHub evidence
CIDM 5310 BUS INTEL & DS SYS	I learned about business intelligence, fact based decision support, ETL, structured and unstructured data, APIs, and the use of AI and LLMs in business. I also learned how data applications can support business strategy.	I can write Python scripts in Google Colab, clean and transform data with Pandas, extract data from APIs, and process network, image, and audio data. I can also build Streamlit applications and connect an LLM API to a chatbot or business workflow.	1. Python data processing 2. Pandas transformations 3. API data extraction 4. Building a Streamlit data application 5. Explaining business uses of AI and analytics	1. Processing image and audio data without examples to follow 2. Diagnosing network data extraction problems 3. Designing reliable LLM prompts and handling weak model responses 4. Managing API authentication, rate limits, and failures independently	GitHub repository
CIDM 6308 SEM DATA/ANALYTICS	I learned about data analytic thinking, research methods, hypothesis testing, data preparation, data visualization, and business intelligence. I also studied supervised and unsupervised learning, regression, decision trees, clustering, and introductory time series analysis.	I can retrieve and prepare data with SQL, Excel, Python, and Google Colab. I can create Power BI visualizations, choose a basic modeling approach for a business question, interpret model output, and communicate the results.	1. Preparing and integrating tabular data 2. Creating Power BI visualizations 3. Distinguishing supervised from unsupervised problems 4. Applying basic regression and classification 5. Translating a business question into an analytics task	1. Checking and explaining all linear and logistic regression assumptions 2. Selecting and tuning models beyond the examples used in class 3. Evaluating clusters when no labeled answer exists 4. Building and validating time series forecasts independently 5. Sustaining an organization wide analytics program	GitHub repository
CIDM 6350 DATA & INFO MGT	I learned relational database concepts, database design, ER and enhanced ER modeling, keys, cardinality, integrity constraints, relational schemas, and normalization. The course also covered transactions, concurrency, security, NoSQL, data lakes, and data warehouses.	I can translate business requirements into ER and EER diagrams, create relational schemas, normalize tables, and write SQL DDL, DML, joins, subqueries, and stored procedures. I also have introductory experience with MongoDB.	1. Creating ER diagrams from requirements 2. Converting models into relational schemas 3. Normalizing tables 4. Writing SQL DDL and DML 5. Querying with joins and subqueries	1. Modeling complex specialization and generalization rules in EER diagrams 2. Reasoning through transaction isolation and concurrency problems 3. Writing stored procedures without reference material 4. Designing MongoDB documents for complex relationships 5. Choosing between a database, data lake, and warehouse for a new system	GitHub repository
CIDM 6355 DATA MINING METHODS	I learned the data mining process from identifying a problem and preparing data through modeling, evaluation, and interpretation. I studied classification, clustering, decision trees, naïve Bayes, logistic regression, SVM, KNN, neural networks, k means, association rules, text mining, model comparison, and data mining ethics.	I can prepare and explore data, build models in R, RStudio, RapidMiner, and Python, and compare classifiers using evaluation measures. I can also perform clustering, association rule analysis, and basic text mining, then explain the results in a business context.	1. Data understanding and preparation 2. Training decision tree and logistic regression models 3. Applying k means clustering 4. Comparing model performance 5. Explaining results in business terms	1. Tuning SVM kernels and parameters without guidance 2. Choosing neural network structures and controlling overfitting 3. Selecting a defensible number of clusters in ambiguous data 4. Interpreting association rule measures beyond basic support and confidence 5. Building a complete text mining process from raw text	GitHub repository
CIDM 6325 WEB APP DEV	I learned about internet and web architecture, client and server responsibilities, front end and back end development, and the Django model view template structure. The course covered URL routing, CRUD, forms, validation, relational models, class based views, authentication, sessions, middleware, testing, deployment, performance, security, accessibility, UX, and business value planning.	I can build Django applications with Python and VS Code, create models and migrations, and implement authenticated CRUD workflows, forms, templates, and class based views. I can use Django admin, write unit tests, measure coverage, manage code with Git and GitHub, and perform a basic deployment and security review.	1. Building Django models, views, templates, and URLs 2. Implementing CRUD workflows 3. Using forms and validation 4. Using Git and GitHub to manage application code 5. Connecting technical features to business value	1. Writing custom middleware without a reference 2. Testing edge cases and maintaining strong coverage 3. Deploying and configuring production settings independently 4. Diagnosing performance problems in a larger Django application 5. Conducting a thorough security audit beyond Django defaults	GitHub repository
CIDM 6303 TECH FOUNDATIONS CIS	I learned programming fundamentals such as input, processing, output, variables, conditionals, loops, functions, objects, and procedural and object oriented design. I also learned about relational databases, SQL, normalization, ER diagrams, HTML, CSS, data science, software engineering ethics, version control, networking, and security.	I can build a basic website with HTML and CSS, write Python programs and notebooks, manipulate data with Pandas, and create visualizations with Seaborn. I can write SQL DDL, DML, and joins, use Google Colab, DataCamp, FreeCodeCamp, Git, and GitHub, and apply basic network security practices.	1. Core Python control flow and functions 2. Data manipulation with Pandas 3. Basic SQL queries and joins 4. HTML and CSS page construction 5. Git and GitHub fundamentals	1. Designing object oriented programs with multiple interacting classes 2. Writing advanced SQL without looking up syntax 3. Creating polished responsive layouts from scratch 4. Resolving complex Git merge conflicts 5. Explaining lower level networking details beyond the introductory material	GitHub repository GitHub repository
CIDM 6340 NTWK MGT & INFO SEC	I learned about organizational networks, common cyberattacks, router security, mobile security, BYOD, Bluetooth security, zero trust, identity and access management, supply chain security, ransomware recovery, and security auditing. I also learned how security strategy relates to risk, ethics, governance, and legal concerns.	I can document and assess a network, discover hosts and services with Nmap, capture and analyze packets with Wireshark, and assess exposure with ShieldsUP! and Nessus. I can use Tableau to present findings, create a ransomware response and recovery plan, and communicate the results of a security audit.	1. Explaining common attacks and basic defenses 2. Running and interpreting basic Nmap scans 3. Filtering and inspecting packets in Wireshark 4. Applying zero trust and least privilege principles 5. Writing a practical ransomware response plan	1. Interpreting unusual packet captures without known good traffic for comparison 2. Distinguishing false positives from actionable Nessus findings 3. Hardening routers, BYOD, and Bluetooth across different vendors 4. Designing enterprise scale IAM and zero trust architecture 5. Performing a full security audit without a checklist or framework	GitHub repository
CIDM 5380 CLOUD COMPUTING & SECURITY	This course is currently in progress during the Fall 2026 semester. Knowledge gained will be assessed and documented upon completion of the course.	To be determined upon completion of the course.	To be determined upon completion of the course.	To be determined upon completion of the course.	GitHub repository